



ASHAR RAIS

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PROFESSIONAL SUMMARY

Dedicated and curious researcher-in-training with a background in biochemistry, cellular biology, and hands-on experience co-founding a medical diagnostics startup. Combines strong academic credentials with technical proficiency in mammalian cell culture, proteomics, and bio-imaging, as well as practical insight into the development and validation of diagnostic solutions. Ready to leverage technical expertise, analytical skills, and entrepreneurial experience to contribute to cutting-edge research in a challenging new project.

EDUCATION

- Master of Science (MS) in Biosciences (specialisation in Cell & Molecular Biology) | King Abdullah University of Science & Technology (KAUST) | 2023-25 | **CGPA: 3.89/4**
- Bachelor of Science (BS) in Biochemistry | Aligarh Muslim University (AMU) | 2021-23 | **CGPA: 8.6/10**

RESEARCH EXPERIENCE

- **Favism’s Overlooked Gene: Examining How a Mutation in CD34 Determines the Severity of G6PD Deficiency** | MS Thesis project | 02/2024-05/2025.
A collaboration project between the Cell Migration & Signalling lab (PI: Prof. Jasmeen Merzaban) of KAUST and Dr. Naif Almontashiri’s lab at Taibah University. This study aims to explore consequences of a CD34 (classical marker of hematopoietic stem cells) mutation upon the functioning of HSCs and how this might affect disease progression in a G6PD patient. With no data till date on the pathogenicity of CD34, this study is the first of its kind and holds great potential in opening new avenues to further understand well-studied blood disorders.

WORK EXPERIENCE

- **Co-Founder & Chief Technology Officer (CTO), HemScan Diagnostics LLP, Aligarh, U.P., India (09/2025 – Present):**
A medical device startup focused on POC blood metabolites analysers reflecting kidney function. As the CTO, my role currently revolves around leading design and development of the first prototype.

TECHNICAL SKILLS

Lab-focussed technical skills:

- Mammalian cell culture;
- Proteomics-based techniques; (Immunoblotting, Immunoprecipitation, Mass Spectrometry and Flow Cytometry);
- RT-qPCR and lentiviral transduction protocols;
- Bio-imaging (brightfield and confocal microscopy);
- Glycosylation-based assays;
- Microfluidics-based cell migration assay;
- Biochemical assays for metabolite detection.

Computational skills:

- Agent-based model (ABM) designing in Python;
- Molecular Dynamic Simulations;

Entrepreneurial skills:

- Viable product design and development;
- Cross-functional leadership.

CERTIFICATES & PUBLICATIONS

- Certificate from American Chemical Society Students Chapter (ACSSC) for an abstract writing competition on the topic “Application of Nanotechnology in Environmental protection” (06/05/2022).
- Review paper: **Rais A**, Husain A, Hasan GM, Hassan MI; A review on regulation of cell cycle by extracellular matrix; International Journal of Biological Macromolecules, Volume 232, 2023, 123426, ISSN 0141-8130. <https://doi.org/10.1016/j.ijbiomac.2023.123426>.
- Review paper (*under review*): Nasif Hussain, Mohd Moin Khan[#], **Ashar Rais[#]**, Nawaz Akhter, Mudassir Meraj Banday, Mohd Mohsin, Asifa Khan and Rizwan Hasan Khan [[#] These authors contributed equally to this work]; Rewriting the Neurodegenerative Code: How Non-coding RNAs Drive Epigenetic and Inflammatory Mechanisms.

LANGUAGE SKILLS

Fluent in:

- English;
- Urdu.

Basic knowledge of:

- Hindi;
- Arabic.

REFERENCES

- Prof. Jasmeen Merzaban | Department of Biomedical Sciences, KAUST | MS Thesis supervisor | jasmeen.merzaban@kaust.edu.sa
- Prof. Christian Froekjaer Jensen | Department of Biomedical Sciences, KAUST | cfjensen@kaust.edu.sa
- Prof. Samah Zeineb Gadhoum | Department of Biomedical Sciences, KAUST | Samah.GADHOUM@kaust.edu.sa

(The information mentioned above is up to date, as of 28/06/2026)